

SPI Buyers' Checklist

CATEGORY	TYPE	DESCRIPTION
Inspection Accuracy	3D/2D Inspection Capability	Mandatory 3D SPI per IPC-7525; automotive traceability; AI-based inspection
	Height Resolution (μm)	$\leq 1\text{-}2\mu\text{m}$ resolution for IPC Class 3 and automotive fine-pitch
	Volume/Area/Height Measurement	Volume $\pm 15\%$, Area $\pm 10\%$; IPC-A-610 compliance; trend monitoring
	Defect Detection Types	Bridging, insufficient, excess, slumping, offset, smear; AI classification
Throughput	Inline/Offline Capability	Inline required for automotive SMT; offline for program creation
	Inspection Speed (Boards/Hour)	$> 800\text{-}1500$ BPH depending on product mix
	PCB Size Support	$50 \times 50\text{mm}$ to $510 \times 510\text{mm}$ as per IPC standards
Integration	Closed Loop to Printer	Real-time SPI-to-printer feedback on alignment, cleaning, pressure
	Line Communication	SMEMA, Hermes, IPC-CFX supported
	SPC	SPC charts Cp/Cpk; statistical alerts; deviation warnings
	MES Connectivity	MES/ERP integration (SECS/GEM, OPC-UA, MQTT)
Flexibility	Fine Pitch/BGA Support	$0.3\text{-}0.4\text{mm}$ pitch compatibility
	Stencil/Paste Variability	Supports multiple paste types; reflective/non-reflective PCB surfaces
Build Quality	Frame Stability	Granite frame or vibration-damping for repeatability
	Repeatability	GR&R $< 10\%$; long-term drift compensation
	Software Usability	Fast job change < 5 min; intuitive UI
Support	Training	IPC-compliant training materials, operator and engineer training
	Warranty	1-year warranty + 3-5 year AMC option
	Spare Parts	48 hr spare availability for automotive uptime
Cost/ROI	Purchase Cost	ROI < 18 months as per automotive expectations
	Maintenance Cost	$< 10\%$ of machine cost annually
	Yield Improvement	Must reduce solder defects by 60-90%
Software Quality	Real-Time Defect Analytics	AI + statistical visualisation; live defect maps
	Auto Threshold Learning	Adaptive tolerance learning based on production trends
	Recipe Management	IPC-2581 support; version control; secured editing
	User Login Control	Role-based access, LDAP/domain login; audit logs
	Cybersecurity	Encrypted communication and data; secure backups
Connectivity	Industry Protocols	SMEMA, Hermes, CFX, SECS/GEM, OPC-UA, MQTT support
	Closed Loop to Pick and Place	Optional feedback to PnP for offset compensation
	Factory Dashboard	Centralised monitoring dashboard across line/machines
	Cloud/Data Storage	SQL + cloud sync; mobile/tablet access
Dashboards	KPI Analytics	FPY, DPU, Cp/Cpk, defect trends, operator performance
	Auto Reports	PDF/Excel scheduled daily/weekly reports
Traceability	PCB Barcode Scanning	2D/QR/Barcode integration; panel/array recognition
	Paste Lot Tracking	Tracks solder paste batch/expiry for each PCB
	Operator ID Logging	Mandatory for automotive IATF audits
	Recipe + Event Logs	All edits, alarms, inspections logged for traceability
	Full Production History	Board-wise history from SPI to reflow correlation

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